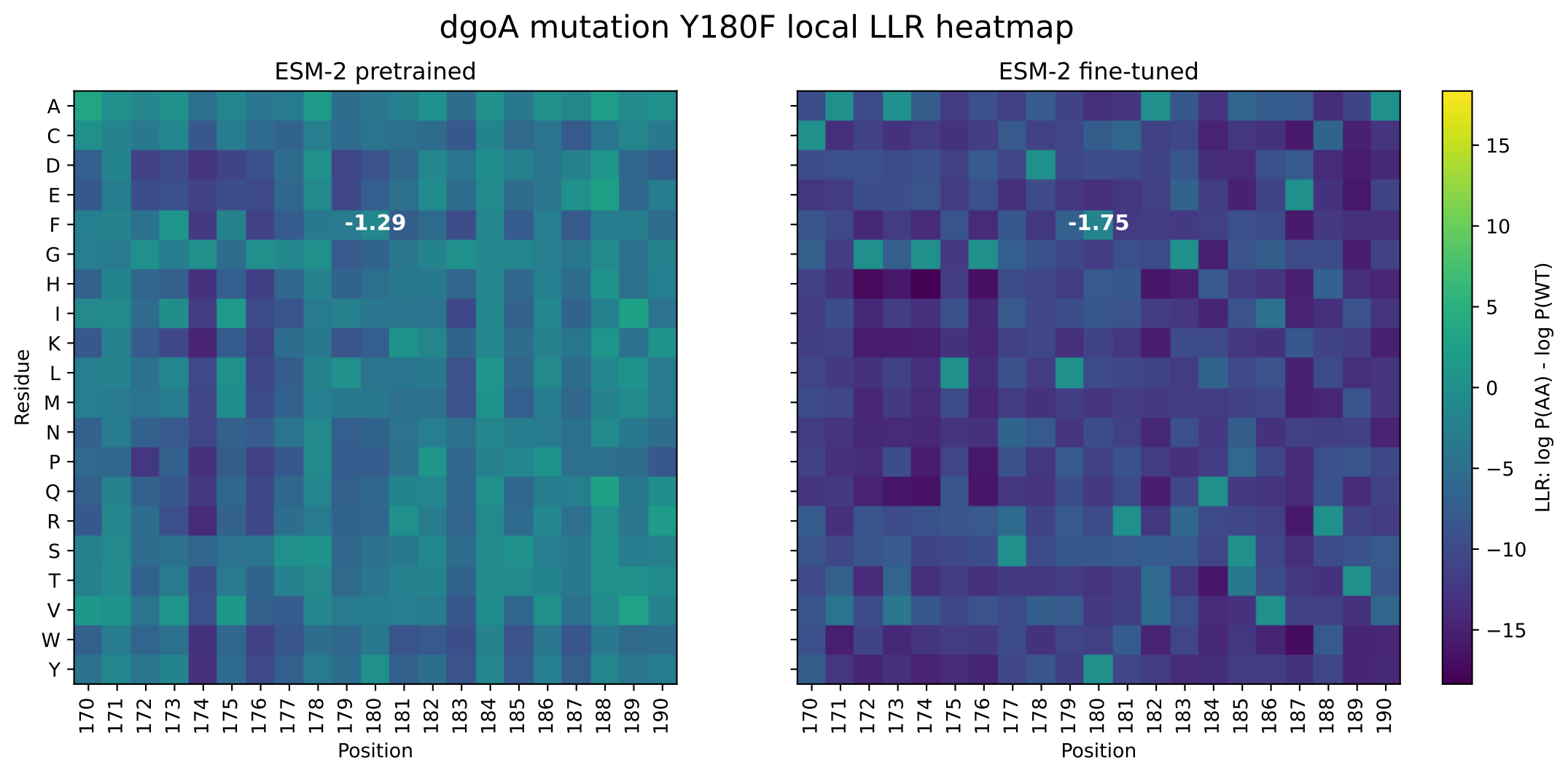
dgoA mutation Y180F local LLR heatmap
ESM-2 pretrained
ESM-2 fine-tuned
-1.29
-1.75
Residue
A
C
D
E
F
G
H
I
K
L
M
N
P
Q
R
S
T
V
W
Y
170
171
172
173
174
175
176
177
178
179
180
181
182
183
184
185
186
187
188
189
190
Position
15
10
5
0
−5
−10
−15
LLR: log P(AA) - log P(WT)